SMS group

Safety provisions for danger zones
Door units and door elements

SN 401-3

ICS 13.110 Dimensions in mm Supersedes SN 401-3: 2010-01

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1 Scope

This company standard specifies the door units and door elements in fence systems according to SN 401-1:2011-08 to be used in the SMS Siemag Business Area of SMS group.

2 Principles

For uniform indication of the dimensions in procurement drawings, the nominal dimensions 1400 and 2200 are used for the height indications of the individual elements.
Unless otherwise specified in other manufacturing documents, the Manufacturing Instructions SN 200 shall be observed for the manufacture of the individual elements.

There are different designs of door elements for door safety category T I and door safety category T II (refer to SN 401-1:2011-08).

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3 Wing door units

Wing door units of door safety categories T I and T II consist of wing door elements, hinge and slamming posts and all other accessories (see tables 1 and 2). They shall always be integrated in the structure as such units with due regard to safety aspects. Wing door units of door safety category T I shall be delivered with the pertaining locks and attachments included in the supply. The lock shall be installed on the site to allow exact adjustment.
Figure 1 shows a wing door unit of safety category T II.

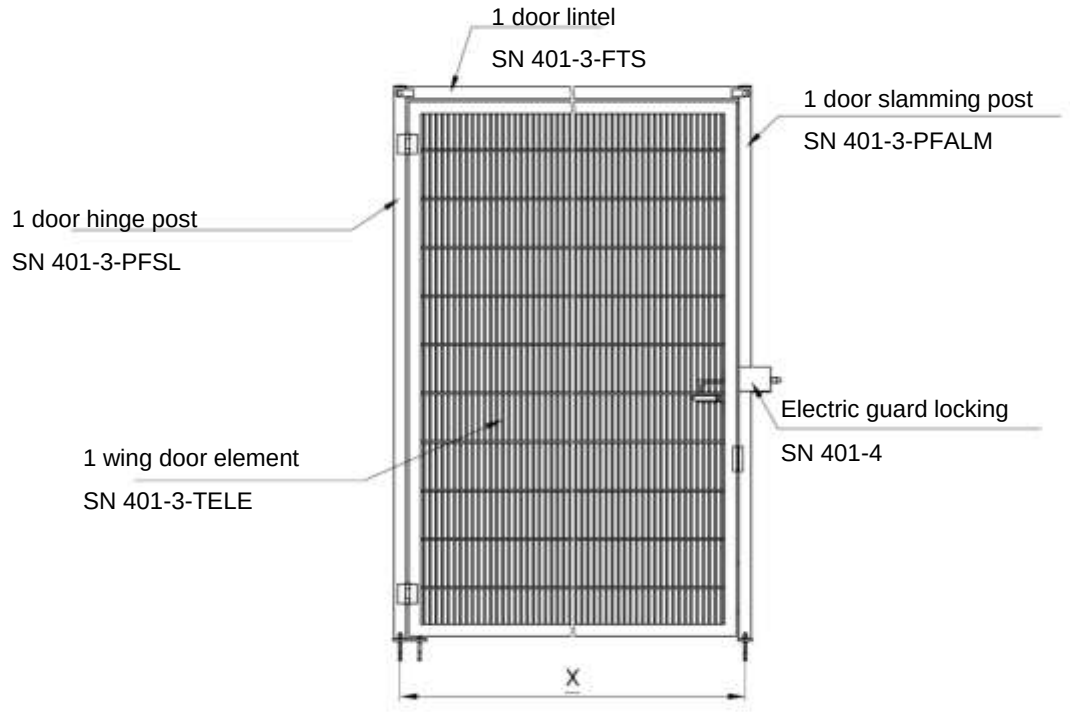


Fig. 1- Wing door unit of door safety category T II (example: FTLE-2200)

Table 1 – Wing door units of door safety category T I / example FTLM – Single elements

Quantity	Designation	Ordering indication	Material indication for fence system made of S 235 JR + N [a)]	Material indication for fence system made of X5CrNi18-10
1	Wing door element	SN 401-3 - TELM	S 235 JR + N	X5CrNi18-10 (1.4301)
1	Door slamming post	SN 401-3 - PFALM		
1	Door hinge post	SN 401-3 - PFSL		
1	Door lintel [b)]	SN 401-3 - FTS		
2	Hexagon head bolt [b)]	ISO 4014 – M10x65	8.8 A0U	A2-70
2	Hexagon nut [b)]	ISO 4032 – M10	8 A0U	A2-70
2	Washer [b)]	ISO 7089 – 10	St-HV200	A2-200

Table 1 – Wing door units of door safety category T I / example FTLM – Single elements (continued)

Quantity	Designation	Ordering indication	Material indication for fence system made of S 235 JR + N [a)]	Material indication for fence system made of X5CrNi18-10
4	Stud anchor	HILTI - HST - M12x115/20	St GLVZ	-
4	Stud anchor	HILTI - HST - HCR - M12x115/20	-	X1NiCrMoCuN25-20-7 (1.4529)
2	Cover cap	SN 401-2 - AK	PE	PE
1	Tubular frame lock	SN 401-3 - RSL	X5CrNi18-10 (1.4301)	X5CrNi18-10 (1.4301)
1	Striking plate (right/left)	SN 401-3 - SB	X5CrNi18-10 (1.4301)	X5CrNi18-10 (1.4301)
1	Profile cylinder, keyed different	SN 401-3 - PZ	Brass	Brass
1	Door latch with long lock plate (inside)	SN 401-3 - TDI	X5CrNi18-10 (1.4301)	X5CrNi18-10 (1.4301)
1	Door latch with long lock plate (outside)	SN 401-3 - TDA	X5CrNi18-10 (1.4301)	X5CrNi18-10 (1.4301)
1	Lock spindle	SN 401-3 - VK	St A0U	St A0U
3	Countersunk head screw	DIN EN ISO 10642 - M 5x60	A2-70	A2-70
5	Blind rivet	DIN EN ISO 15978 - 4,8x10-AlA/St-H	St/Al	St/Al

[a)] Fence system made of S235JR+N (with subsequent powder coating) or S235JR+N TZN
[b)] Only for wing door units with a nominal height of 2200 mm.

Table 2 – Wing door units of door safety category T II / example FTLE – Single elements

Quantity	Designation	Ordering indication	Material indication for fence system made of S 235 JR + N [a)]	Material indication for fence system made of X5CrNi18-10
1	Wing door element	SN 401-3 - TELE	S 235 JR + N	X5CrNi18-10 (1.4301)
1	Door slamming post	SN 401-3 - PFALE		
1	Door hinge post	SN 401-3 - PFSL		
1	Door lintel [b)]	SN 401-3 - FTS		
2	Hexagon head bolt [b)]	ISO 4014 – M10x65	8.8 A0U	A2-70
2	Hexagon nut [b)]	ISO 4032 – M10	8 A0U	A2-70
2	Washer [b)]	ISO 7089 – 10	St-HV200	A2-200
4	Stud anchors	HILTI - HST - M12x115/20	St GLVZ	-
4	Stud anchors	HILTI - HST-HCR - M12x115/20	-	X1NiCrMoCuN25-20-7 (1.4529)
2	Cover caps	SN 401-3 - AK	PE	PE

[a)] Fence system made of S235JR+N (with subsequent powder coating) or S235JR+N TZN
[b)] Only for wing door units with a nominal height of 2200 mm.

3.1 Design types of wing doors, design requirements

Wing door units as defined in this standard are available in two design types (see figures 2, 3 and table 3).
For reasons of safety aspects, only such doors may be used in a protective fence system which open in the direction away from the danger zone (escape doors).

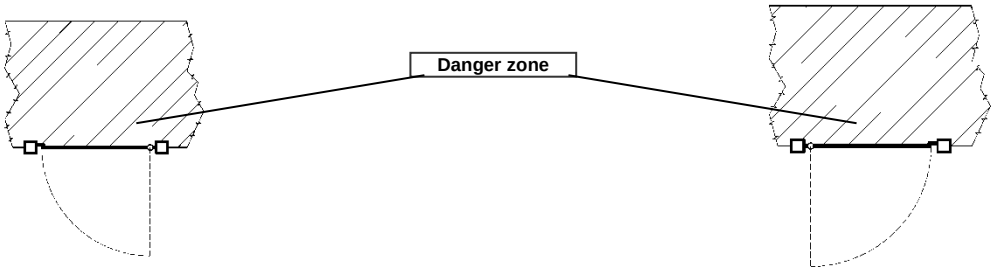
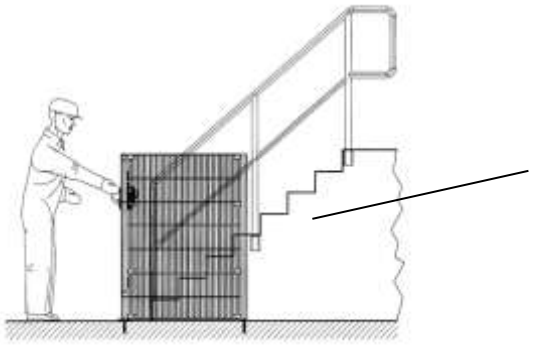


Fig. 2 – Wing door unit FTR
(hinged on the right, opening to the outside)

Fig. 3 – Wing door unit FTL
(hinged on the left, opening to the outside)

Fence systems shall be designed in compliance with the following requirements:

- The wing door unit shall be without any connection to the surrounding fence system.
- Fence nettings shall never be fastened direct to a door post, but only to a separate fence post.
- Door units shall not be installed at the very corner of a fencing as this makes it impossible to comply with the specified gap dimensions. There shall be at least one fence netting between a fence corner and the door unit.
- To prevent any foreseeable bypassing of the protective fence system, door units in front of staircases shall be provided with additional lateral fence elements (see Fig. 4).
- Posts installed near each other shall be positioned in such a way that the distances between the foot plates (welding item 002 in figures 12 to 17) do not exceed 10 mm.



Example of lateral protection with additional fence element.
Without additional fence element the safety door can be bypassed as the guard rail of the stairway does not prevent access to the danger zone.

Fig. 4 – Additional lateral fence element

Table 3 – Wing door units – Design types

Ordering indication	Door safety category	Nominal height	Clear opening width B	Drillhole distance X [a)]	Weight kg/unit	Opening direction
SN 401-3 - FTRM	T I	1400	950	1020	54,15	right
			1350	1420	58,58	
		2200	950	1020	81,57	
			1350	1420	93,05	
SN 401-3 - FTLM		1400	950	1020	54,15	left
			1350	1420	58,58	
		2200	950	1020	76,12	
			1350	1420	85,42	
SN 401-3 - FTRE	T II	1400	950	1020	51,05	right
			1350	1420	56,58	
		2200	950	1020	78,47	
			1350	1420	89,95	
SN 401-3 - FTLE		1400	950	1020	51,05	left
			1350	1420	56,58	
		2200	950	1020	78,47	
			1350	1420	89,95	

[a)] see dimension X in figure 1.

3.2 Hinge bands

The wing door elements and the hinge posts shall be levelled and aligned together by the supplier for the welding on of the hinge bands. The inner half, J (see Fig. 5), shall be welded up with the door element, the outer half, A, with the hinge post. After this, the surface treatment can be carried out. If door element and posts are shipped separately afterwards, the parts belonging together shall be unmistakably match-marked by the supplier. Table 4 shows the dimensions of the hinge band.

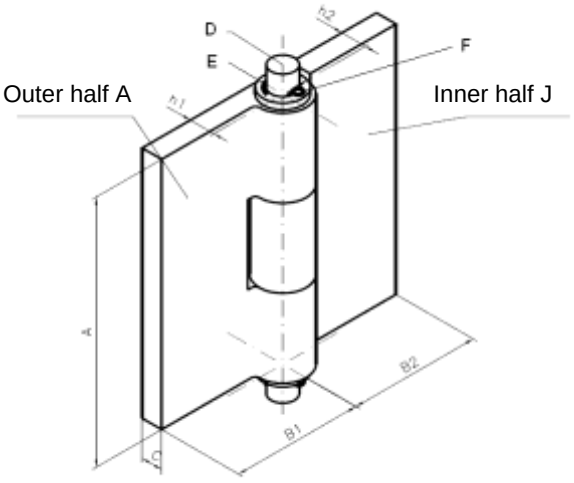


Fig. 5 - Hinge band GB

Hinge band SN-401-3-GB
(developed from DIN 31211; has been withdrawn)

Table 4 – Dimensions

A	80 mm
B1	40 mm
B2	40 mm
C	7 mm
h1	8 mm
h2	8 mm
D	Clevis pin DIN EN 22340-8x90
E	Washer DIN EN ISO 7089
F	Split pin DIN EN ISO 1234-2x15

Hinge bands and their connecting parts shall be ordered to suit the material selected for the fence and door elements.

4 Door elements

Door element is the general term for all individual elements used to make up wing door units and their posts (see Fig. 1).

4.1 Wing door elements

Depending on their use for door units of door safety category T I or T II, wing door elements shall be ordered in different design types. The types TERM and TELM are provided with preparation for the installation of a mechanical lock, the types TERE and TELE have no such preparation. Table 5 states the dimensions of the wing door elements. Figures 6 to 9 show the elements of wing doors.

Table 5 – Wing door elements

Ordering indication	Door safety category	Height H	Clear opening width B	Weight kg/unit	Opening direction
SN 401-3 - TERM	T I	1400	950	31,90	right
			1350	39,43	
		2200	950	44,85	
			1350	54,15	
SN 401-3 - TELM		1400	950	31,90	left
			1350	39,43	
		2200	950	44,85	
			1350	54,15	
SN 401-3 - TERE	T II	1400	950	35,66	right
			1350	43,19	
		2200	950	50,69	
			1350	59,99	
SN 401-3 - TELE		1400	950	35,66	left
			1350	43,19	
		2200	950	50,69	
			1350	59,99	

4.1.1 Wind door element TERM

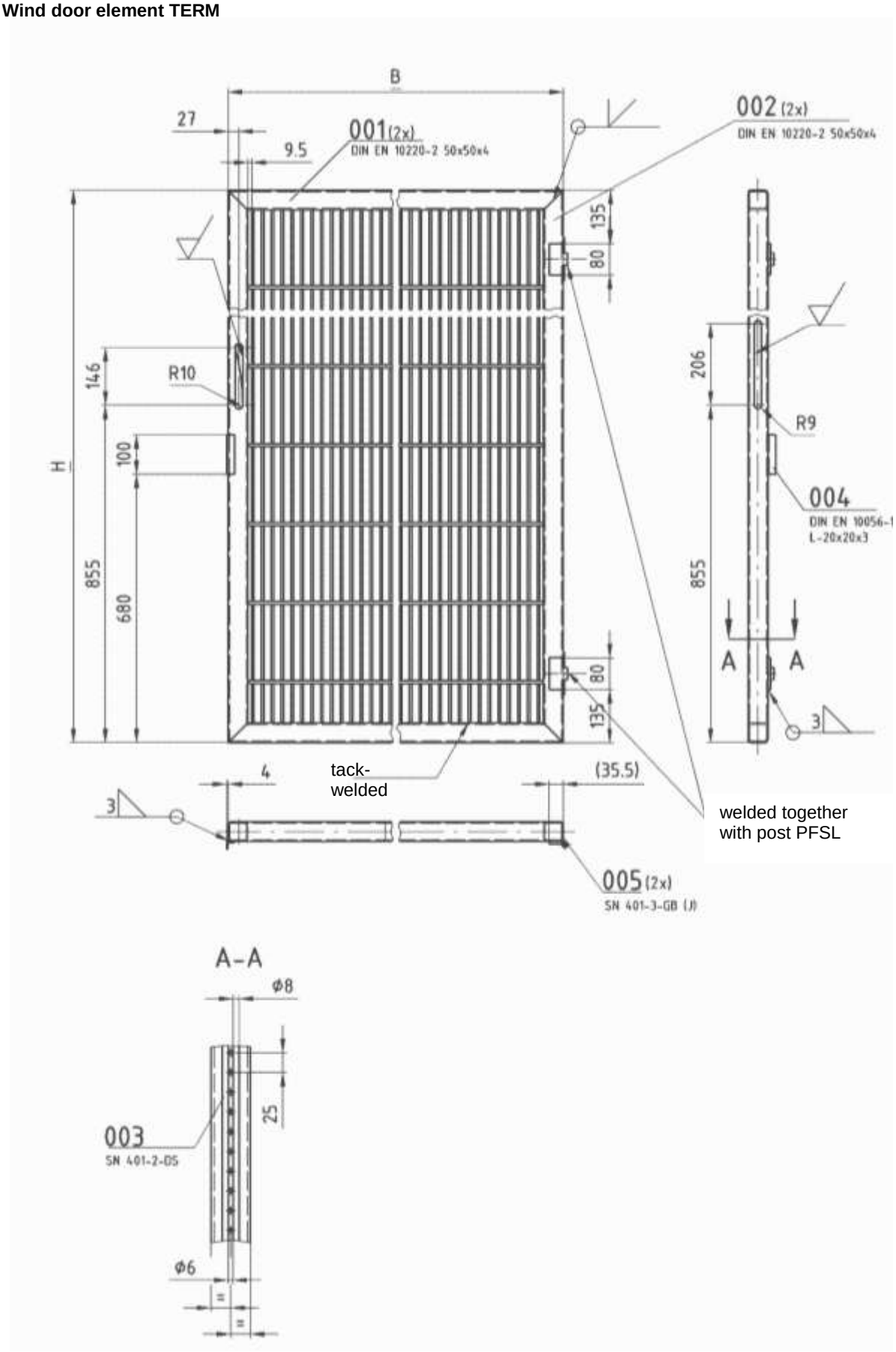


Fig. 6 – Wing door element TERM

4.1.2 Wing door element TELM

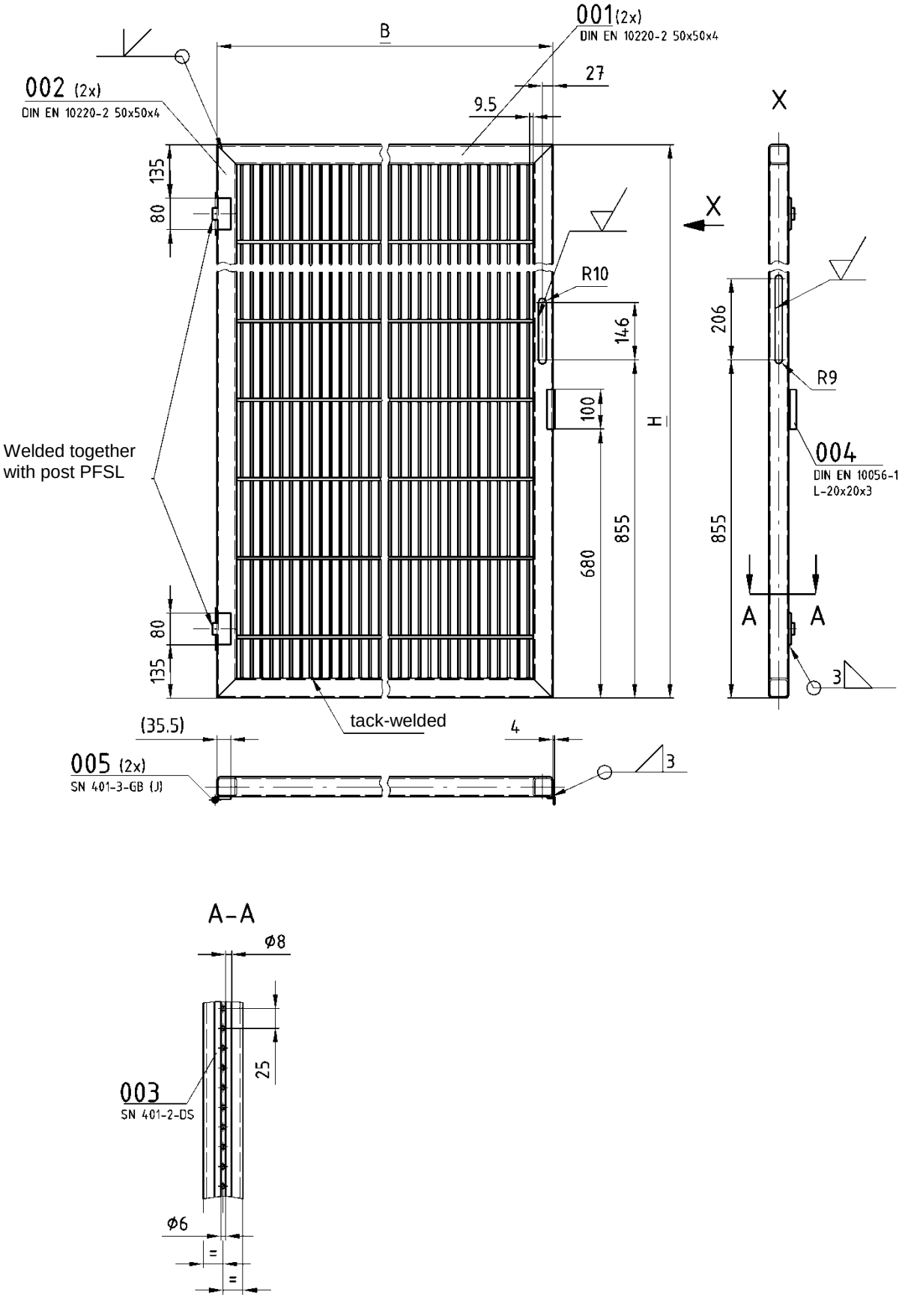


Fig. 7 – Wing door element TELM

4.1.3 Wing door element TERE

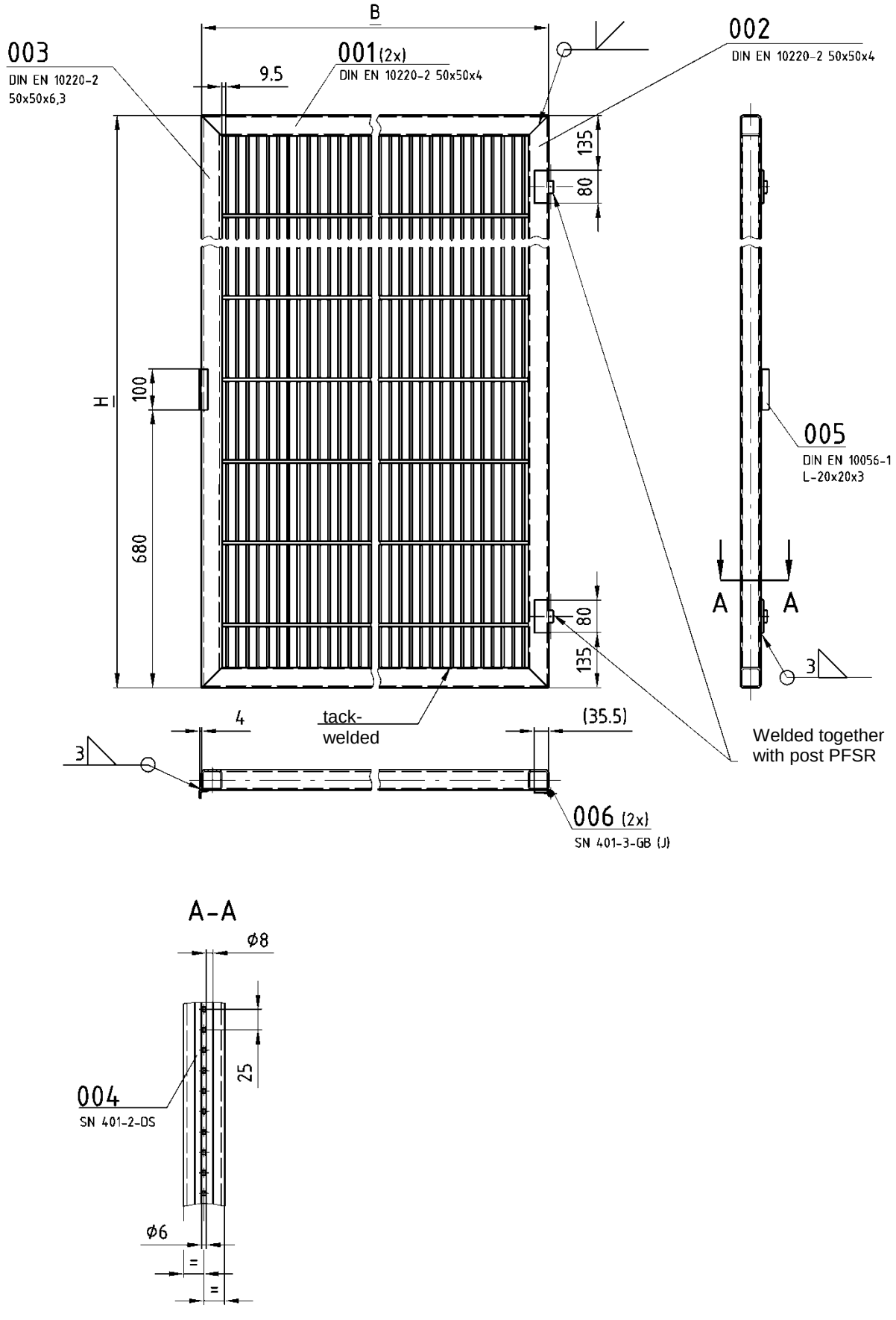


Fig. 8 – Wing door element TERE

4.1.4 Wing door element TELE

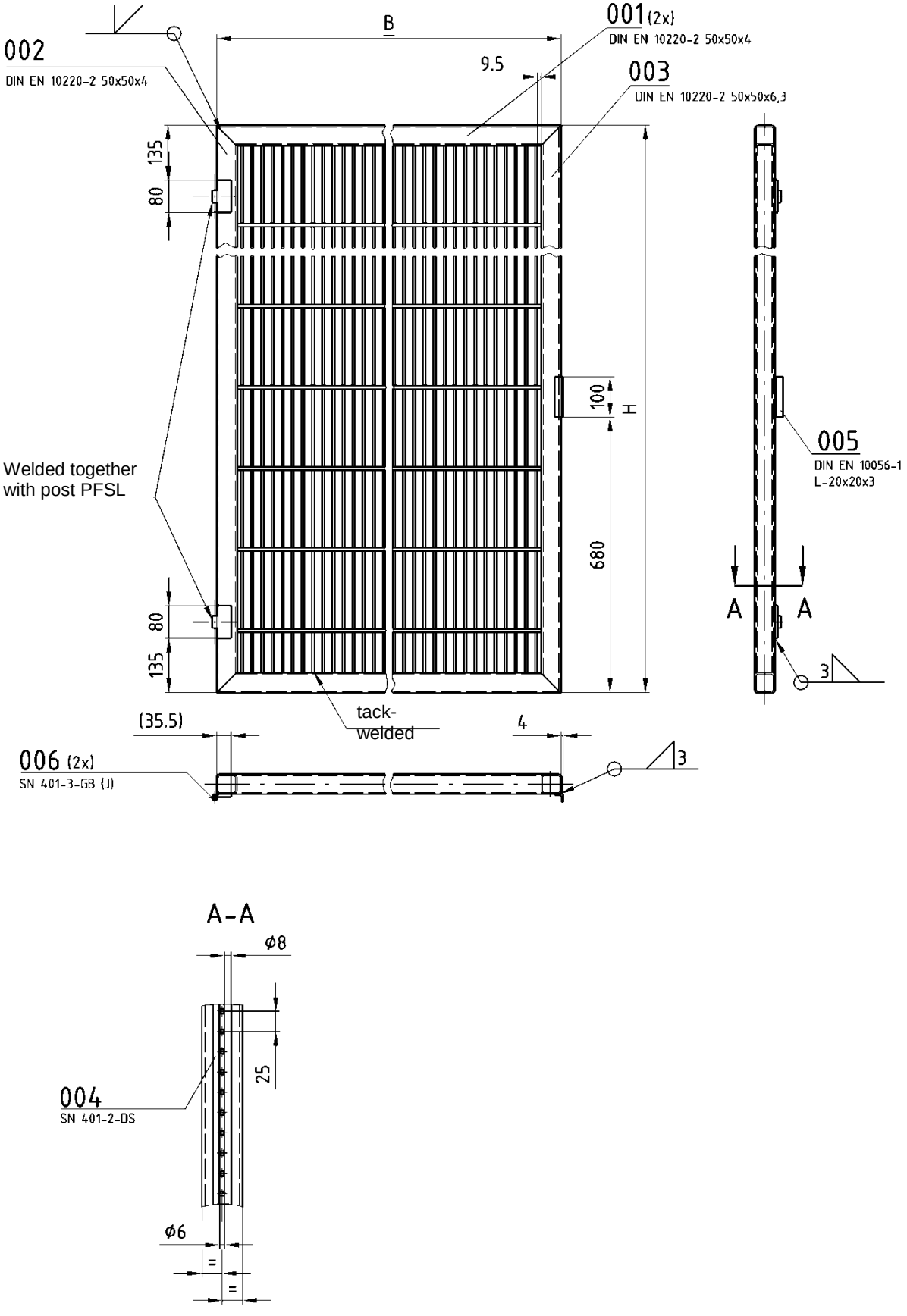


Fig. 9 – Wing door element TELE

4.2 Wing door post

To avoid unintentional locking in of persons in the danger zone, an angle section with oblong hole shall be provided on the slamming post of the wing door for the use of a personal padlock (Figs. 10 and 11). This personal padlock is intended to show that one or more persons are still staying in the danger zone. In addition, the padlock prevents the closing of the door.
The slamming posts and hinge posts are listed in Table 6 and illustrated in the figures 12 to 17.

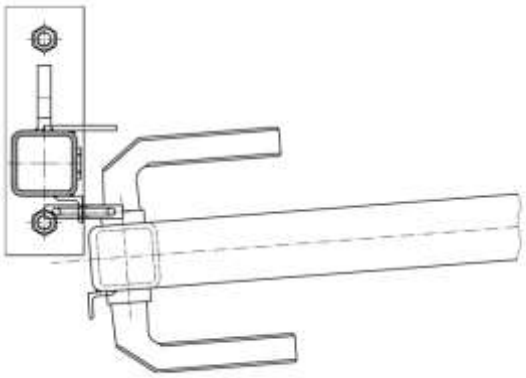


Fig. 10 – Device for personal padlock with door open

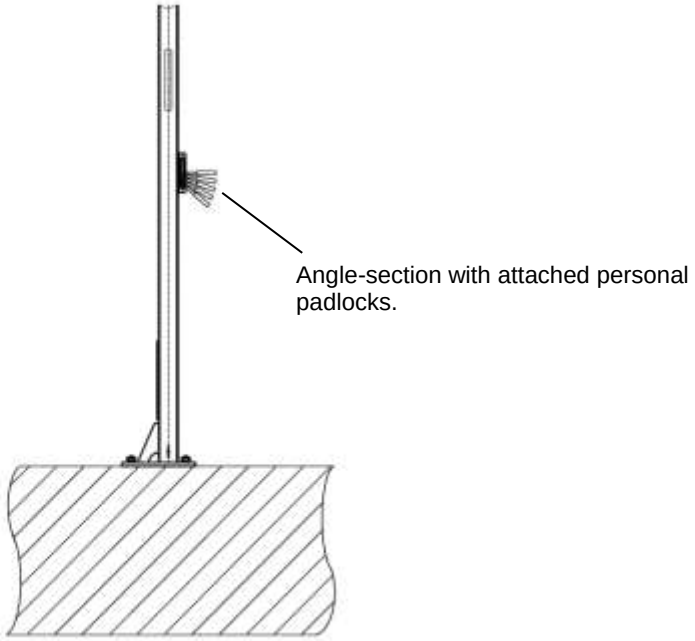


Fig. 11 – Device for personal padlock at slamming post

Table 6 – Wing door posts

Ordering indication	Function	Door safety category	Height H	Weight kg/unit	Opening direction
SN 401-3-PFARM-1400	Slamming-post	T I	1410	9,35	right
SN 401-3-PFARM-2200			2270	14,24	
SN 401-3-PFARE-1400		T II	1410	12,99	
SN 401-3-PFARE-2200			2270	19,96	
SN 401-3-PFALM-1400		T I	1410	9,35	left
SN 401-3-PFALM-2200			2270	14,24	
SN 401-3-PFALE-1400		T II	1410	12,99	
SN 401-3-PFALE-2200			2270	19,96	
SN 401-3-PFSR-1400	Hinge post	T I + T II	1410	10,13	right
SN 401-3-PFSR-2200			2270	15,03	
SN 401-3-PFSL-1400			1410	10,13	left
SN 401-3-PFSL-2200			2270	15,03	

4.2.1 Wing door slamming post PFARM

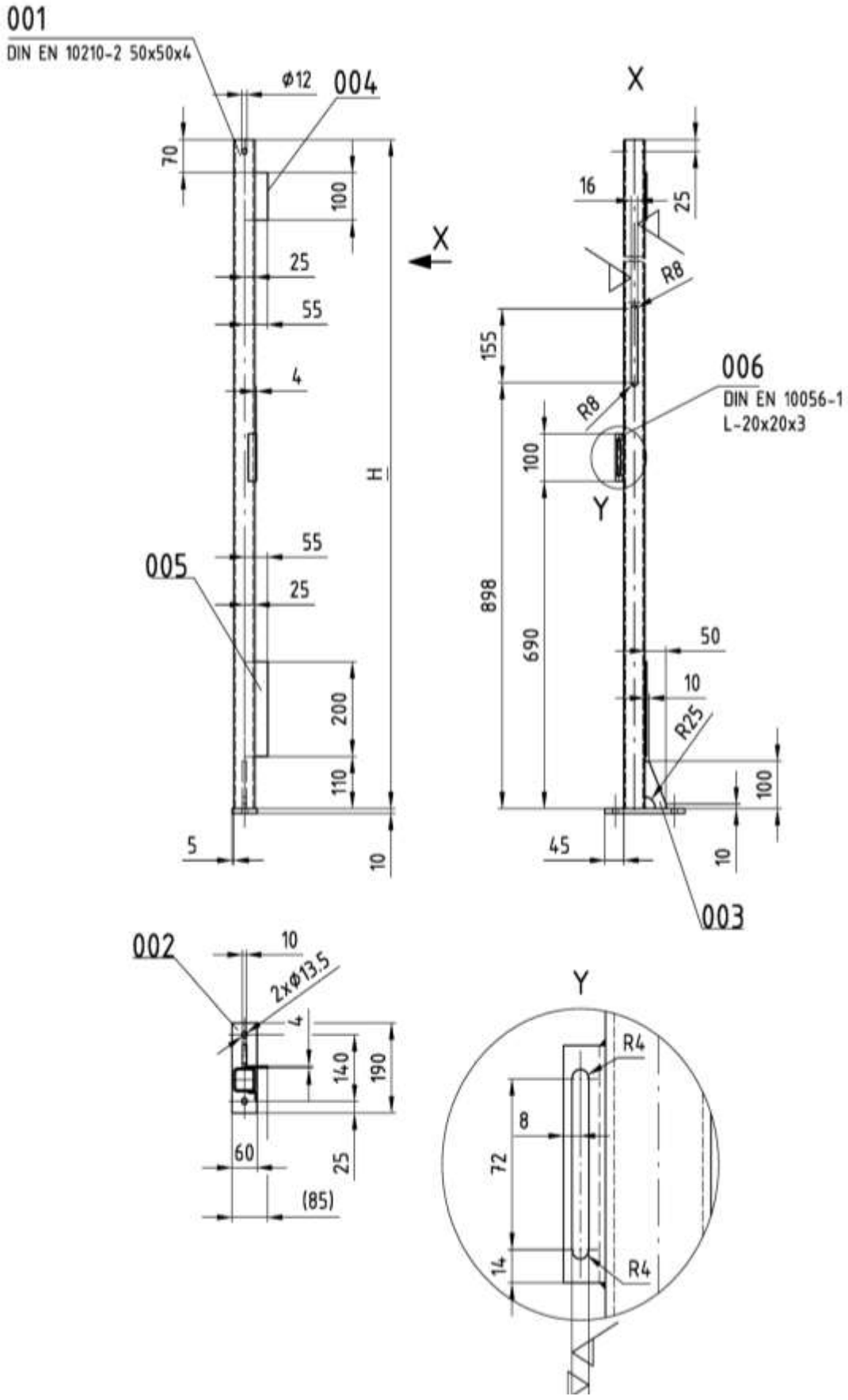


Fig. 12 – Wing door slamming post PFARM

4.2.2 Wing door slamming post PFARE

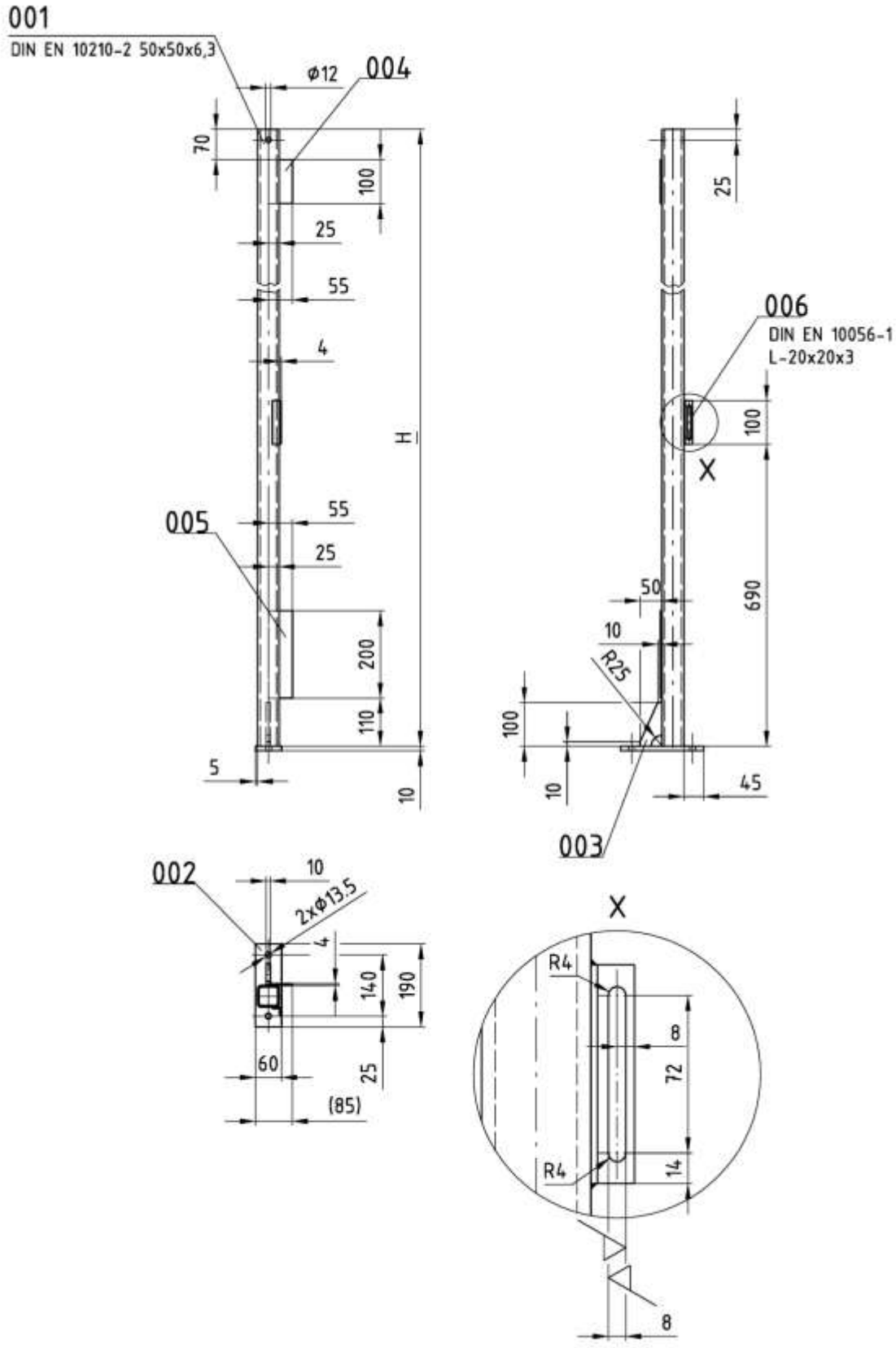


Fig. 13 – Wing door slamming post PFARE

4.2.3 Wing door slamming post PFALM

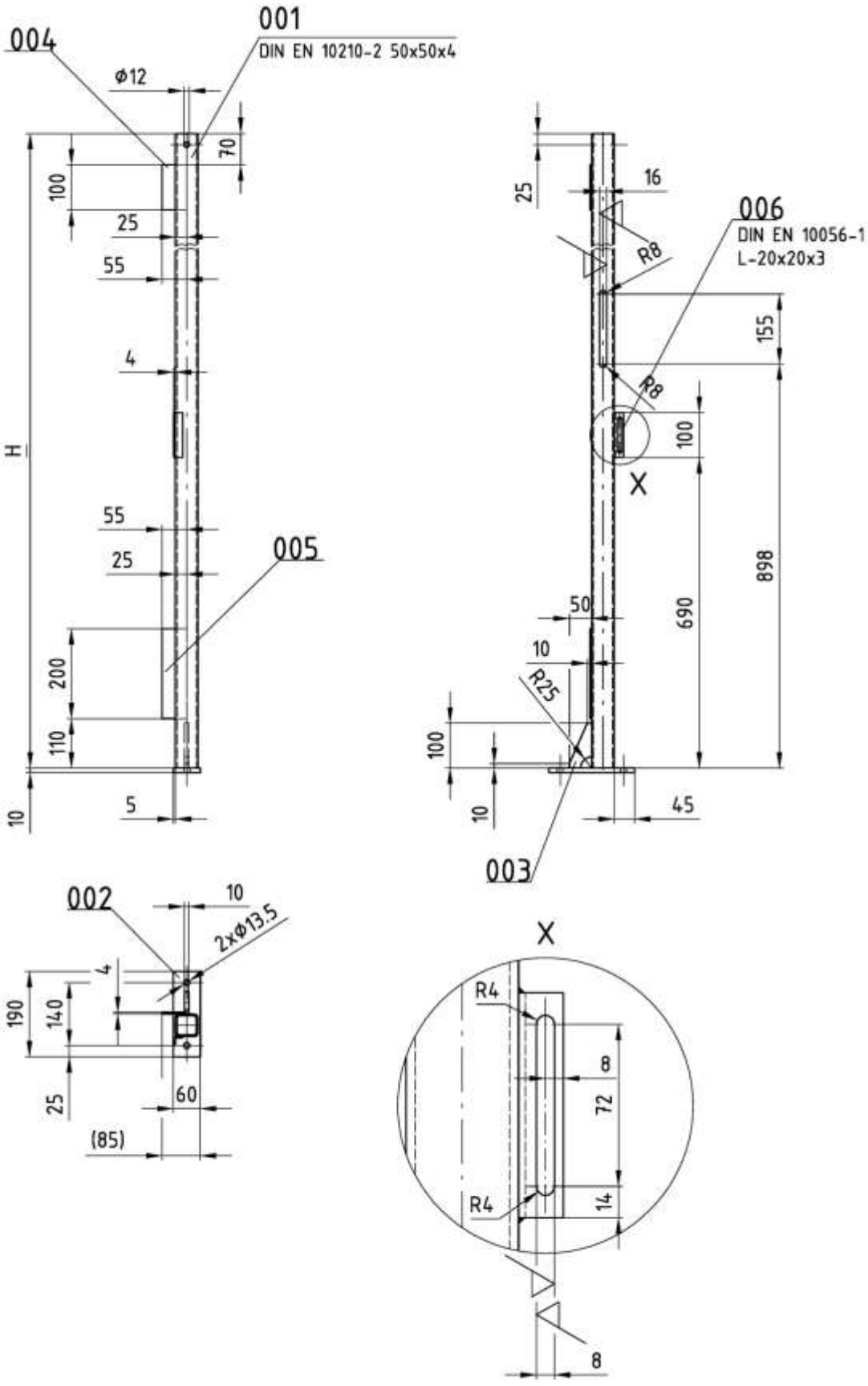


Fig. 14 – Wing door slamming post PFALM

4.2.4 Wing door slamming post PFALE

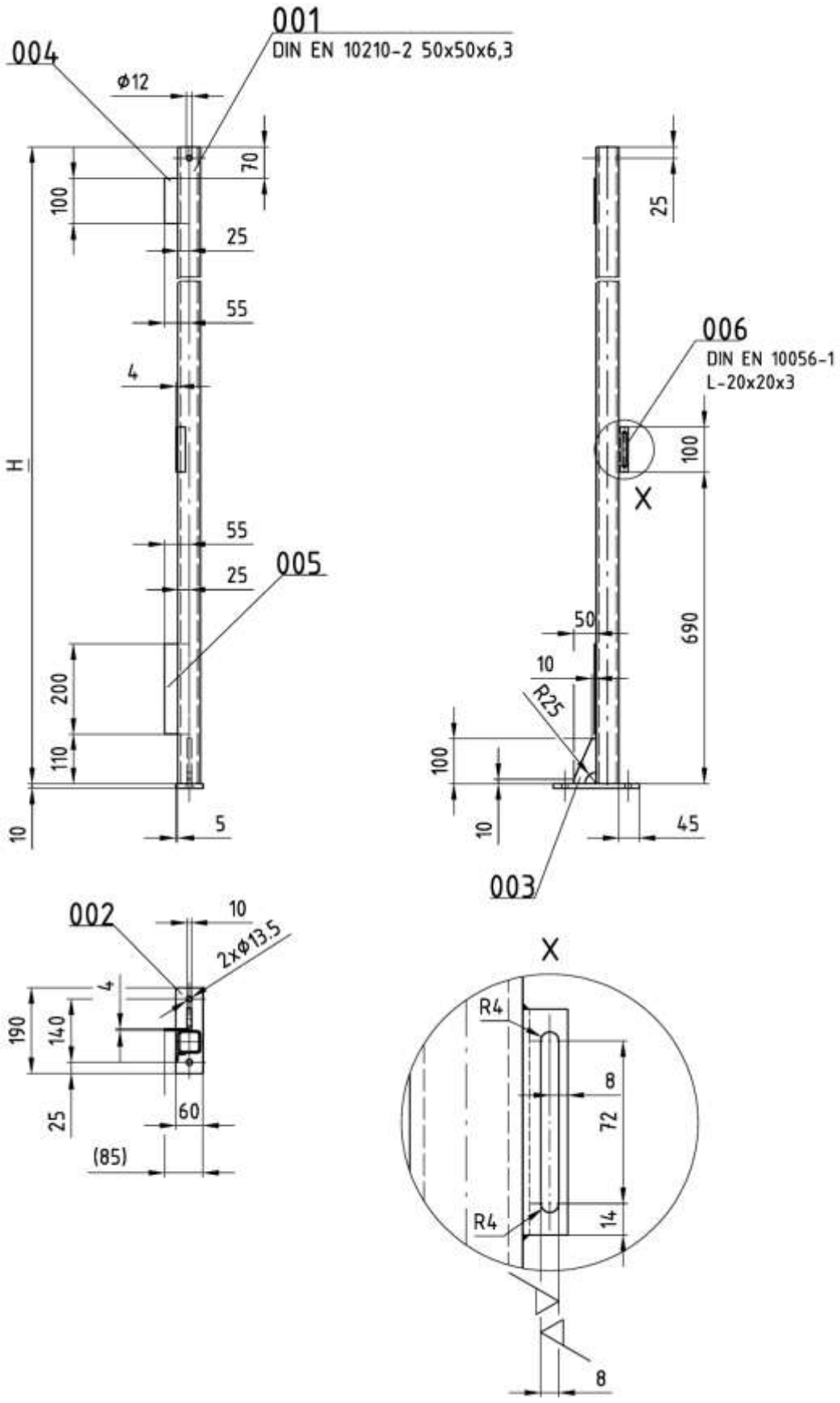


Fig. 15 – Wing door slamming post PFALE

4.2.5 Wing door hinge post PFSR

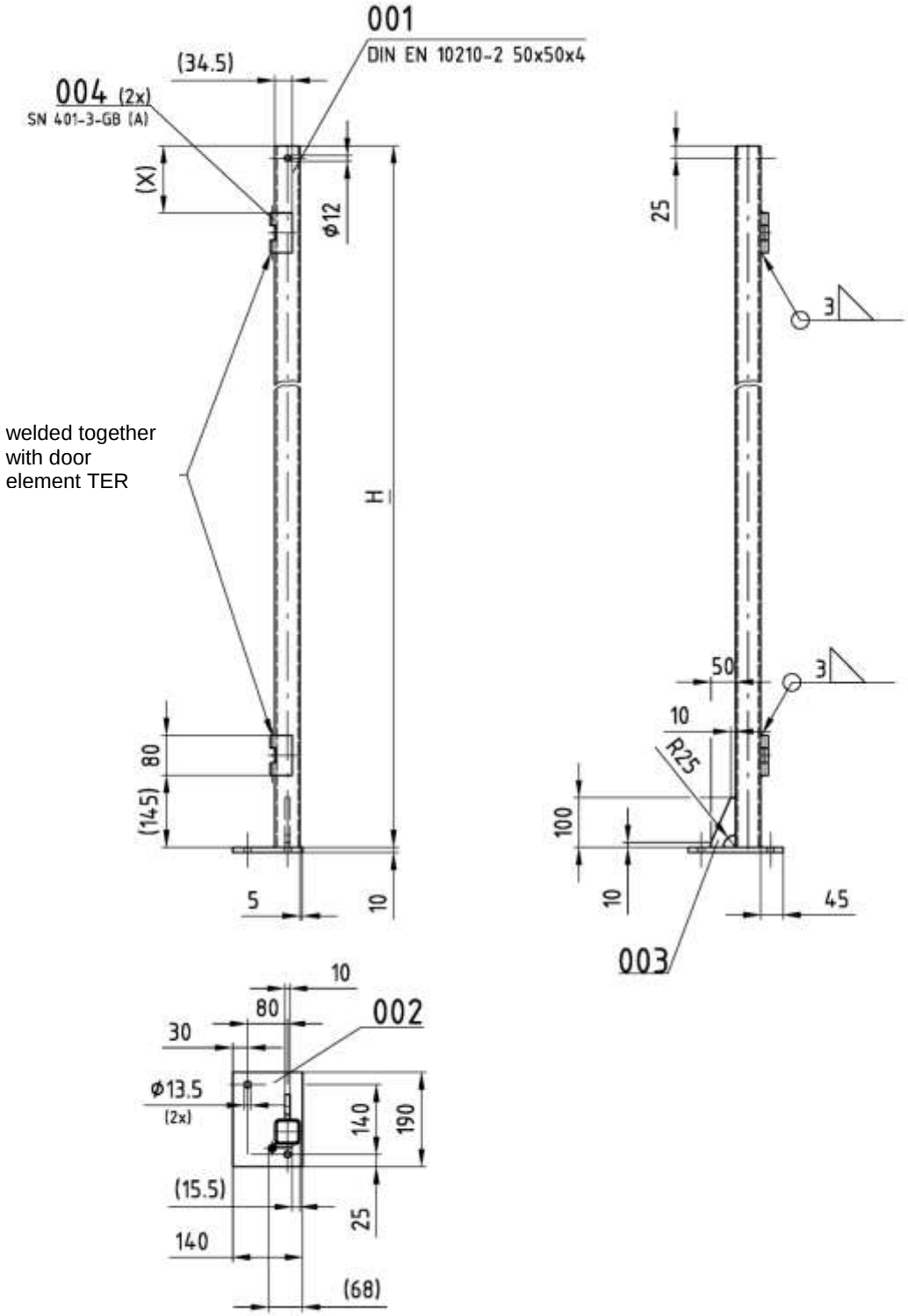


Fig. 16 – Wing door hinge post PFSR

4.2.6 Wing door hinge post PFSR

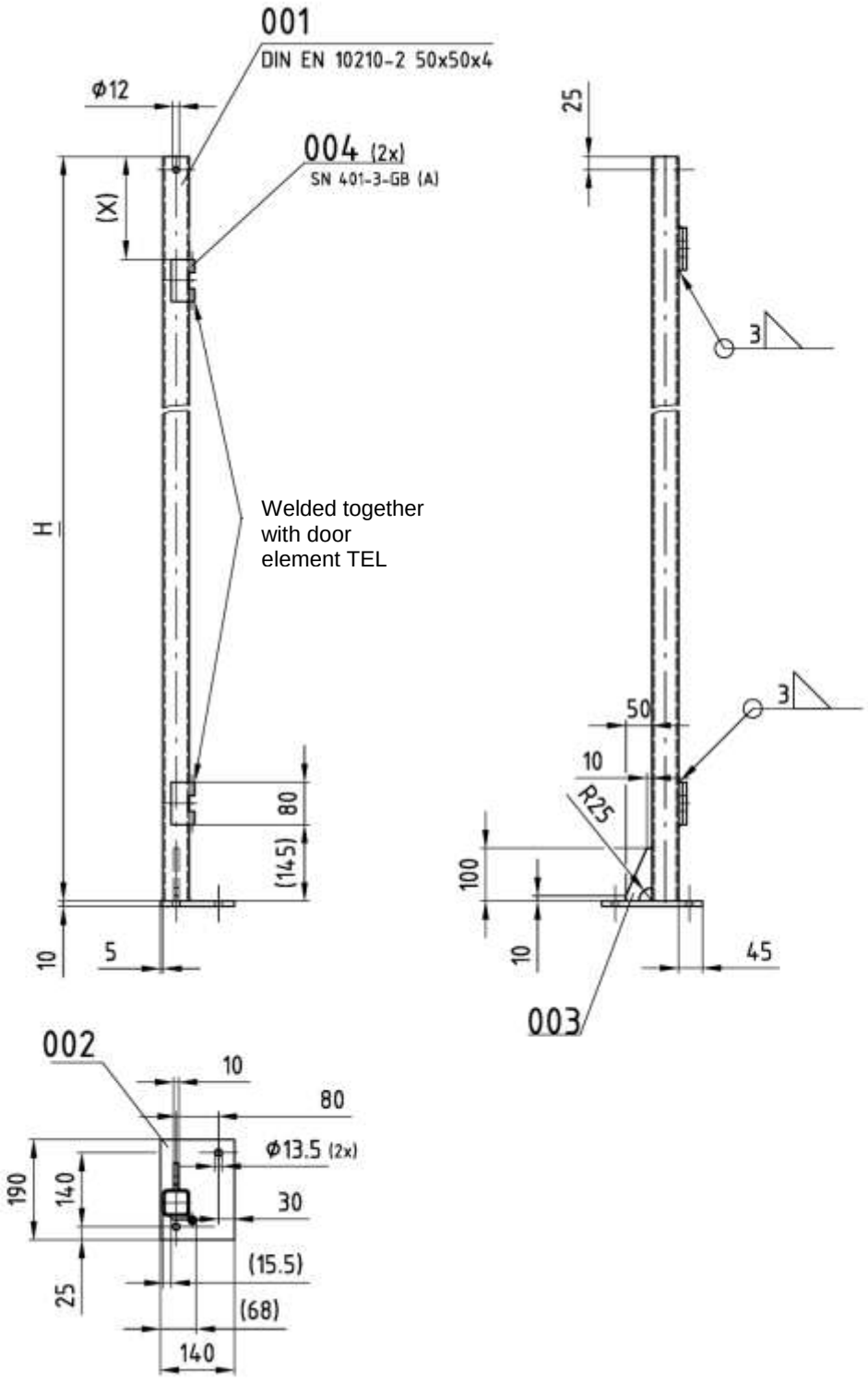


Fig. 17 – Wing door hinge post PFSL

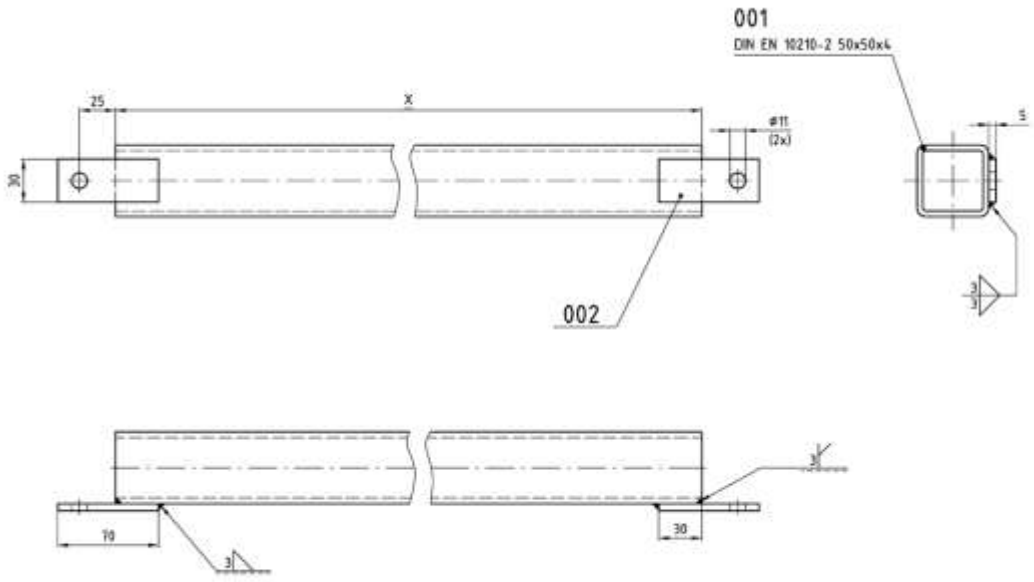


Fig. 18 – Door lintel FTS

Table 7 – Door lintels

Ordering indication	Door safety category	Clear opening width B	Length X	Weight kg/unit	Opening direction
SN 401-3 – FTS-950	T I + T II	950	970	5,46	right and left
SN 401-3 – FTS-1350	T I + T II	1350	1370	7,64	right and left

5 Variants of door locks

In fence systems of door safety category T I, all door units stated in this standard are provided with the appropriate tubular frame lock and accessories. The individual elements are shown in Table 7.
For doors of door safety category T II, an appropriate door lock shall be determined in accordance with the requirements and the type of door lock variant shall be specified in addition to the door unit. The electric components required for door safety category T II are specified in SN 401-4.:2010-03.
The safety devices required in the respective door safety category shall be applied at all door units of a danger zone.

5.1 Door lock, mechanical

The mechanical locking of door units according to this SN 401-3 makes it necessary to use door elements of the types TERM or TELM, the types TERE and TELE are not provided with preparation for the installation of mechanical locks (electric guard locking or key transfer system).
The door locks shall be installed on the customer's site to ensure exact adjustment.
The tubular frame lock and the striking plate are fastened to the door and the door post with open end blind rivets with countersunk head according to DIN EN ISO 15978.
Mechanical door locks shall be keyed different, i.e. every lock shall have its own key. Every locking cylinder shall be supplied by the manufacturer with at least two keys.
The individual elements are illustrated in figures 19 to 24.

5.1.1 Tubular frame lock RS

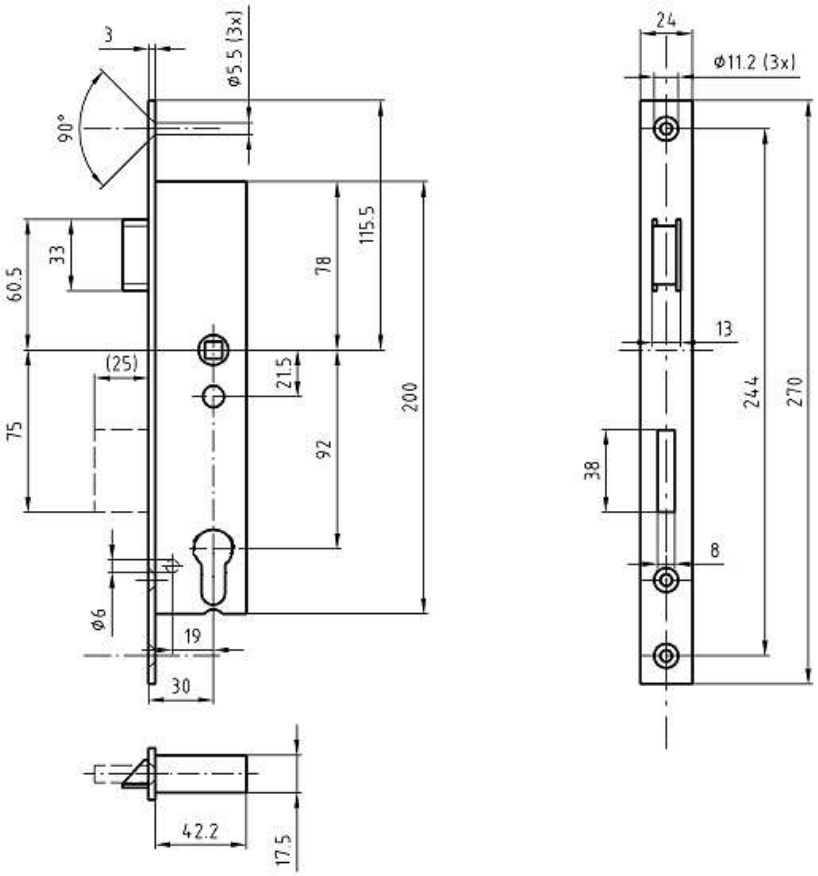


Fig. 19 – Tubular frame lock RSR (example: right hinged)

5.1.2 Striking plate SB

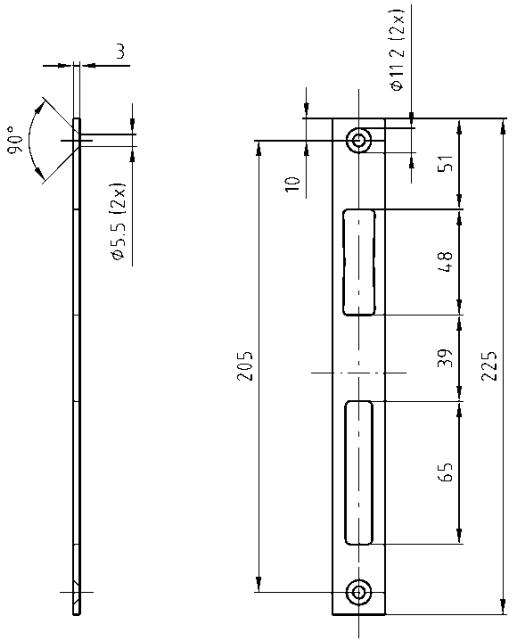


Fig. 20 – Striking plate SB

5.1.3 Profile cylinder PZ

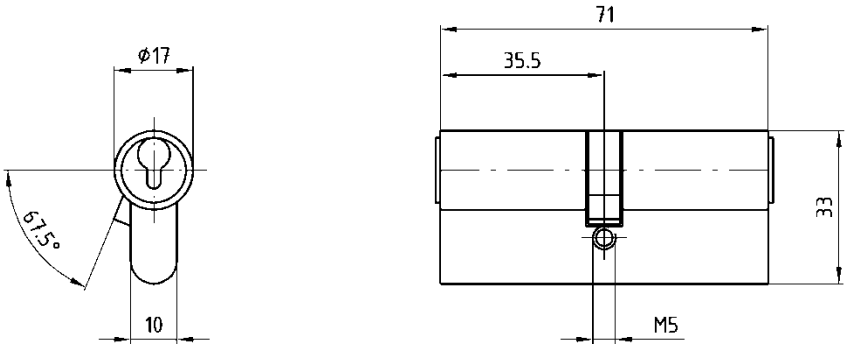


Fig. 21 – Profile cylinder PZ

5.1.4 Door latch TDA

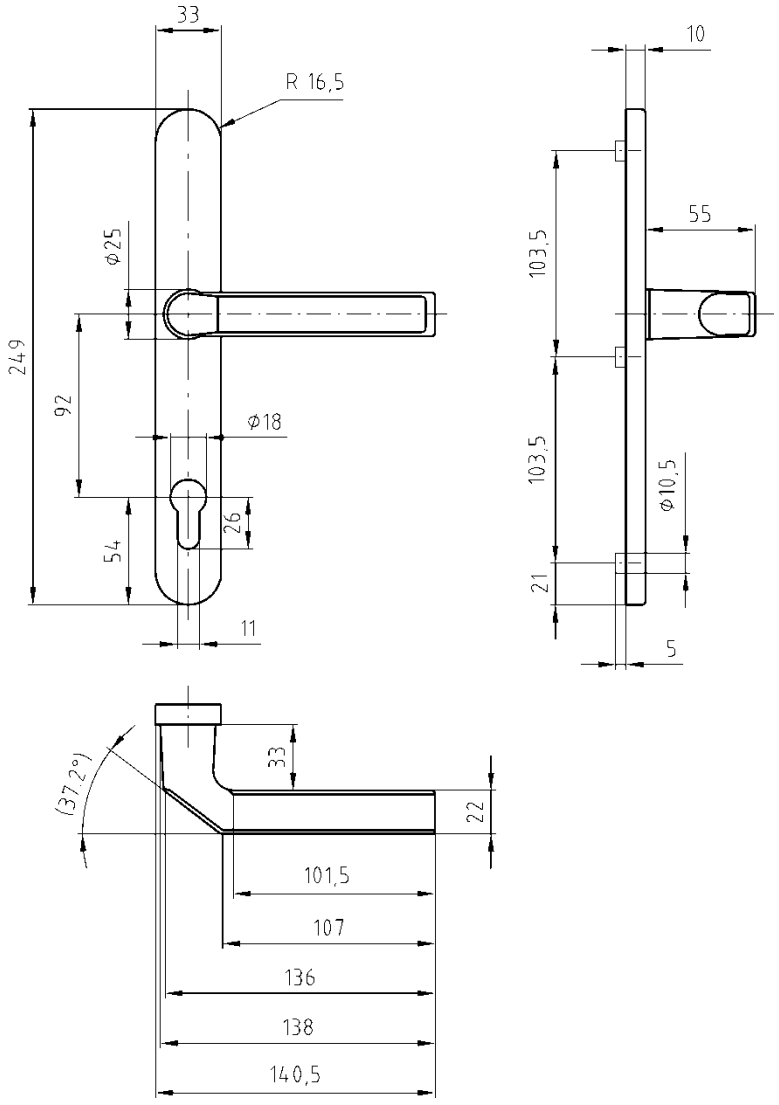


Fig. 22 – Door latch TDA

5.1.5 Door latch TDI

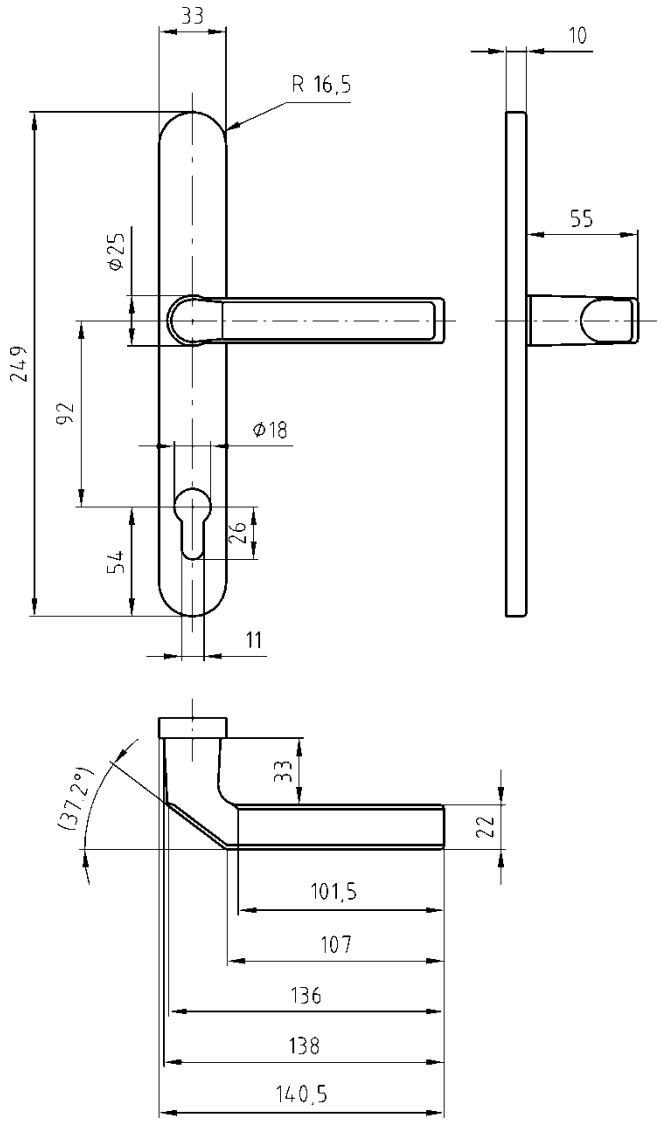


Fig. 23 – Door latch TDI

5.1.6 Lock spindle VK

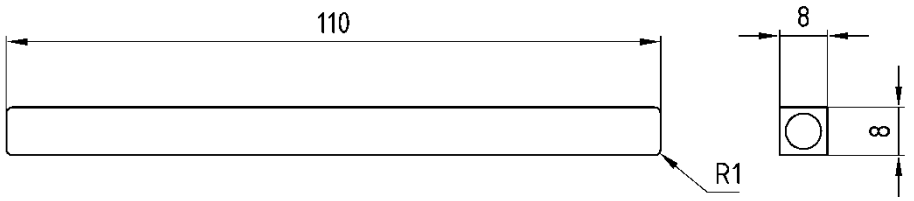


Fig. 24 – Lock spindle VK

6 Materials

A2-70	according to DIN EN ISO 3506-1
A2-200	according to DIN EN ISO 3506-1
Brass	brass, general, non-specific
PE	according to DIN EN ISO 1043-1, polyethylene
S235JR	according to DIN EN 10025-2:2005-04, structural steel, surface untreated (for subsequent powder coating)
S235JR TZN	according to DIN EN 10025-2:2005-04, structural steel, surface galvanised
St/Al	steel/aluminium
ST GLVZ	steel, electrogalvanised
ST A0U	steel, surface galvanised
X1NiCrMoCuN25-20-7 (1.4529)	according to DIN EN 10028-7:2008-02, austenitic stainless steel, without surface treatment
X5CrNi18-10 (1.4301)	according to DIN EN 10088-3:2005-09, austenitic stainless steel, without surface treatment

7 Examples of designation

Designation of a wing door unit of nominal height 1400 mm, 950 mm clear opening width, left hinged, door safety category T II, made of galvanised structural steel:

Wing door unit SN 401-3-FTLE-1400x950-S235JR-TZN

Designation of a door hinge post of 2200 mm nominal height, for right hinged doors, made of X5CrNi18-10 (1.4301):

Door hinge post SN 401-3-PFSR-2200-X5CrNi18-10

Designation of a door lock profile cylinder:

Door lock profile cylinder SN 401-3-PZ

Referenced technical standards, codes and regulations

DIN 31211 (withdrawn)	Hinges; Assemblies of parts
DIN EN 3506-1	Mechanical properties of corrosion-resistant stainless steel fasteners - Part 1: Bolts, screws and studs
DIN EN 10025-2:2005-04	Hot rolled products of structural steels - Part 2: General technical delivery conditions for non-alloy structural steels
DIN EN 10028-7:2008-02	Flat products made of steels for pressure purposes - Part 7: Stainless steels
DIN EN 10056-1	Structural steel equal and unequal leg angles - Part 1: Dimensions
DIN EN 10088-3:2005-09	Stainless steels - Part 3: Technical delivery conditions for semi-finished products, bars, rods, wire, sections and bright products of corrosion resisting steels for general purposes
DIN EN 10210-2	Hot finished structural hollow sections of non-alloy and fine grain steels - Part 2: Tolerances, dimensions and sectional properties
DIN EN 22340	Clevis pins without head
DIN EN ISO 1043-1	Plastics - Symbols and abbreviated terms - Part 1: Basic polymers and their special characteristics
DIN EN ISO 1234	Split pins
DIN EN ISO 7089	Plain washers - Normal series, product grade A
DIN EN ISO 10642	Hexagon socket countersunk head screws
DIN EN ISO 15978	Open end blind rivets with break pull mandrel and countersunk head – AIA/St
SN 200	Manufacturing instructions
SN 401-1:2011-08	Safety provisions for danger zones; Fence systems; Overview - Definitions
SN 401-4:2010-03	Safety provisions for danger zones; Electric door interlocking - Design directive

Revision of April 2013

Section on Principles newly added.
Tables 1, 2, 3, 5 and 7 modified.
Figures revised and newly added.